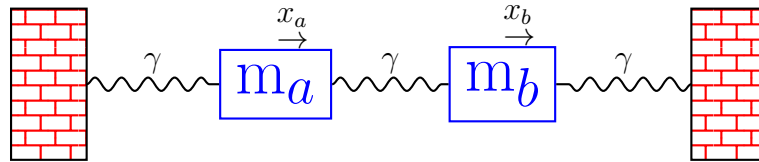


Homework 3

Implication of normal modes, symmetry, etc

1. Consider the two mass problem yet again.

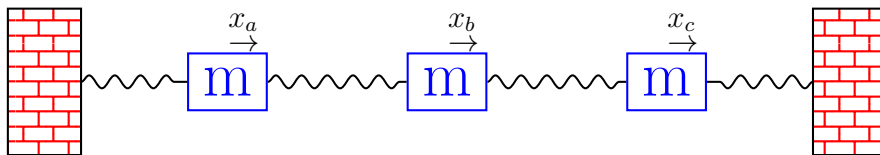


- (a) Assume that $m_a = m_b = m$, and that the initial conditions are $x_a = x_o$, $x_b = 2x_o$, $v_a = v_b = 0$. Sketch the “phase portrait” (i.e. plot the momentum as a function of the position) for both the even and the odd normal mode. Please correctly label the intercepts on the position and momentum axis for each curve.
- (b) Once again assume that $m_a = m_b = m$, and consider the partition function:

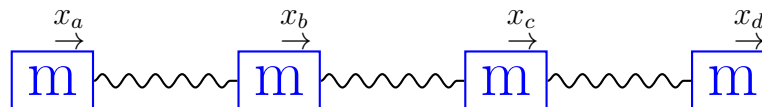
$$Z = \frac{1}{h} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} dp_a dp_b dx_a dx_b e^{-\beta H}$$

Please compute the partition function and also evaluate the internal energy.

2. Consider the three mass problem in 1d with equal masses. Assume that potential energy respects mirror symmetry about the center particle.



- (a) Write the stiffness matrix in the naive basis (i.e. x_a, x_b, x_c) using only irreducible derivatives as dictated by the mirror and Symmetric nature of stiffness matrix.
 - (b) Assume that I choose one of the odd vectors to be $\frac{1}{\sqrt{10}}(|X_a\rangle + 3|X_b\rangle + |X_c\rangle)$. Please find the orthonormal odd vector. You may want to use Gram-Schmidt.
3. Consider the four mass problem in 1d with a mirror in the center.



Please construct a new, complete basis consisting of Eigenvectors of the mirror.

4. Consider the 1d chain with translation symmetry for $N = 2$ (i.e. 2 sites).
- (a) Write out the Eigenvectors of the translation operations. Explicitly operate on them with all translations to confirm they are correct.
 - (b) Construct the stiffness matrix in the naive basis in terms of irreducible derivatives, accounting for the stiffness matrix being symmetric, translation symmetry, and the acoustic sum rule.
 - (c) Determine the Eigenvalues.